

LAPPEENRANTA, Finland

WMO: 029580

Lat: 61.044N

Lon: 28.151E

Elev: 349

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-5.5	-16.0	2.3	-11.2	-9.5	3.3	-5.3	18.0	28.5	16.4	27.6	4.6	10	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	81.2	65.5	77.9	63.6	74.7	62.0	67.7	77.3	65.8	74.2	63.9	71.8	7.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	92.0	71.5	62.5	85.8	69.5	60.7	80.4	67.4	32.3	77.4	30.8	74.2	29.3	72.0	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.9	13.4	DB	-14.7	84.9	6.8	4.3	-19.6	88.0	-23.6	90.5	-27.4	92.9	-32.3	96.0	(4)
(5)			WB	-15.1	69.9	6.7	2.4	-19.9	71.7	-23.9	73.1	-27.7	74.4	-32.5	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	18.9	19.2	26.9	39.0	50.1	58.1	64.3	60.7	51.7	40.3	31.5	25.2		
	DBStd	17.68	12.23	11.90	8.26	7.09	7.84	6.60	5.85	5.46	5.91	6.83	7.89	10.41		
	HDD50	4670	965	863	717	340	96	8	0	1	51	306	555	768		
	HDD65	9009	1430	1283	1182	781	465	225	85	155	399	766	1005	1233		
	CDD50	1246	0	0	0	9	99	252	444	334	102	6	0	0		
	CDD65	107	0	0	0	0	3	19	63	22	0	0	0	0		
	CDH74	825	0	0	0	0	39	155	480	150	1	0	0	0		
	CDH80	152	0	0	0	0	4	21	105	22	0	0	0	0		
	WSAvg	6.5	6.5	6.4	6.7	6.3	6.7	6.6	6.0	5.9	6.2	6.6	7.0	7.3		
	PrecAvg	25.4	1.8	1.4	1.5	1.3	1.4	2.1	2.7	3.1	2.6	2.6	2.6	2.2		
	PrecMax	37.8	4.4	4.2	2.9	3.0	4.4	5.1	5.9	6.2	5.6	5.6	5.9	5.2		
(15) Precipitation	PrecMin	17.3	0.4	0.2	0.2	0.2	0.2	0.2	0.0	0.0	0.4	0.6	0.4	0.4		
	PrecStd	4.4	0.9	0.8	0.7	0.7	0.7	1.1	1.4	1.6	1.3	1.4	1.1	1.1		
	DB	37.8	39.3	49.4	66.5	78.7	82.8	86.6	82.6	70.3	57.4	48.6	45.2			
	MCWB	36.9	36.2	40.5	51.8	59.7	64.5	67.3	66.0	60.1	52.7	47.0	43.8			
	2% DB	36.2	36.1	43.2	61.0	73.1	78.7	82.8	78.3	66.5	54.0	46.1	41.3			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	35.6	34.8	37.1	49.1	58.1	63.0	66.6	63.6	58.4	51.8	45.1	40.4			
	5% DB	34.5	34.5	39.5	56.0	68.5	74.8	79.3	74.5	63.7	52.1	44.2	39.2			
	MCWB	34.0	33.8	35.0	46.0	54.8	60.6	65.3	62.5	57.0	50.1	43.4	38.5			
	10% DB	32.9	33.1	36.6	51.3	64.5	70.7	75.8	71.0	60.9	50.1	42.0	37.0			
	MCWB	32.6	32.6	33.8	42.9	52.4	59.1	63.6	61.3	55.9	48.7	41.3	36.4			
	0.4% WB	37.5	36.7	41.5	55.2	63.5	67.8	70.9	69.0	62.1	54.5	47.3	43.9			
	MCDB	37.7	37.8	48.1	63.2	74.6	76.8	81.1	78.5	66.6	56.0	48.0	45.0			
	2% WB	35.7	35.2	38.1	50.4	59.8	65.2	68.7	66.0	60.1	52.6	45.1	40.6			
	MCDB	36.1	36.0	41.9	58.7	68.5	74.7	78.7	73.5	64.4	54.0	45.8	41.1			
	5% WB	34.1	33.9	35.9	46.8	56.9	62.8	67.1	64.2	58.3	50.8	43.2	38.3			
(33) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	34.4	34.4	38.4	54.4	65.6	71.6	76.3	71.7	62.1	51.9	43.8	38.9			
	10% WB	32.4	32.5	34.3	43.7	53.9	60.2	65.2	62.4	56.6	48.8	41.3	36.2			
	MCDB	32.6	32.9	36.0	49.7	63.0	68.6	73.4	69.0	60.2	49.9	41.8	36.7			
	MDBR	8.9	9.9	12.3	15.0	17.2	15.7	15.2	14.6	12.5	8.9	6.9	7.7			
	5% DB	6.6	7.6	15.4	22.2	23.4	21.2	19.9	19.7	16.2	10.8	8.0	6.9			
(36) Mean Daily Temperature Range	MCWBR	6.7	7.0	10.5	12.5	11.6	9.6	8.0	8.4	9.2	8.2	7.7	6.9			
	5% WB	6.5	7.0	12.3	20.3	20.1	18.5	18.1	16.9	14.3	9.6	7.7	6.6			
	MCWBR	6.6	6.7	8.9	12.1	11.3	9.2	8.3	8.5	9.0	8.0	7.5	6.7			
	taub	0.256	0.298	0.323	0.362	0.358	0.355	0.379	0.366	0.341	0.323	0.280	N/A			
	taud	2.075	2.135	2.213	2.340	2.429	2.468	2.418	2.443	2.500	2.400	2.163	N/A			
(42) Clear-Sky Solar Irradiance	Ebn at Noon	146	211	250	264	275	278	266	260	247	207	135	N/A			
	Edh at Noon	13	23	30	32	32	31	32	29	23	18	12	N/A			
	RadAvg	76	260	725	1181	1615	1695	1664	1280	749	304	82	40			
(45) All-Sky Solar Radiation	RadStd	15	59	95	127	185	111	157	146	83	44	13	5			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-391	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	-322	-340	N/A	N/A

Nomenclature: See separate page